

Problema No. 1

a) 1er. Buete

$$1. i = \frac{n}{2} + 0 \Rightarrow i = \frac{n}{2} + k - 1 = n \Rightarrow k = \frac{n}{2} + 1 \therefore \Theta(n)$$

$$2. i = \frac{n}{2} + 1$$

$$3. i = \frac{n}{2} + 2$$

2do Buete

$$1. j = 1 + \frac{n}{2} \Rightarrow j = k + \frac{n}{2} = n \Rightarrow k = \frac{n}{2} \therefore \Theta(n)$$

$$2. j = 2 + \frac{n}{2}$$

$$3. j = 3 + \frac{n}{2}$$

3er Buete

$$1. k = 1 \Rightarrow k = 2^{p-1} = n \Rightarrow p-1 = \log_2(n) \Rightarrow p = \log_2(n) + 1 \therefore \Theta(\log(n))$$

$$2. k = 2$$

$$3. k = 4$$

$$T(n) = \sum_{i=0}^n O(n) \cdot O(n) \cdot O(\log n) = O(n^2 \log n) \sum_{i=0}^n \frac{1}{2^i} = O(n^2 \log n) \cdot 2 = O(n^2 \log n)$$

$$= \sum_{i=0}^n \left[\frac{n}{2^i} + n - \log_2(n) \right] = \sum_{i=0}^n \frac{n}{2^i} - \log_2(n) \cdot (n+1) = \left(\frac{n}{2} + n + n + \dots + n \right) - \log_2(n) \cdot (n+1) = \frac{n}{2} \cdot 2 - \log_2(n) \cdot (n+1) = n - \log_2(n) \cdot (n+1) = O(n)$$

$$\frac{n}{2} + \frac{n}{2} + \frac{n}{2} + \dots + \frac{n}{2} = \frac{n}{2} \cdot 2 = n$$